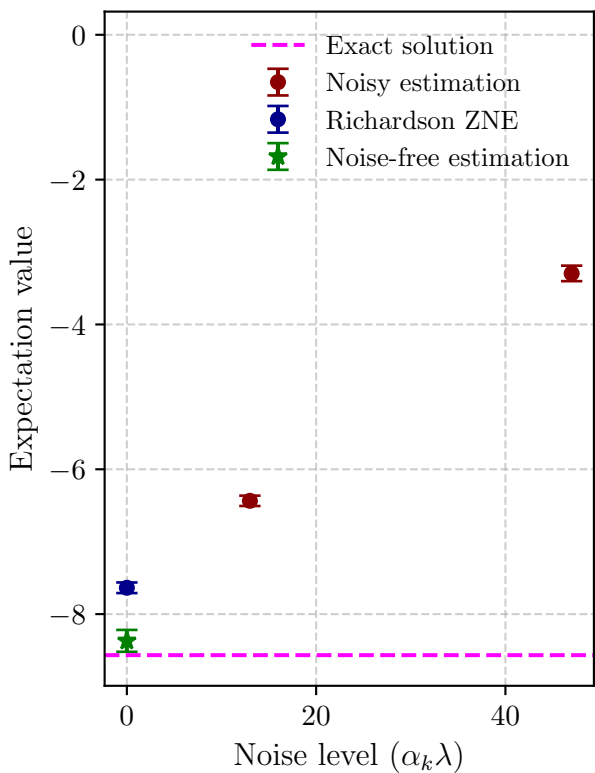
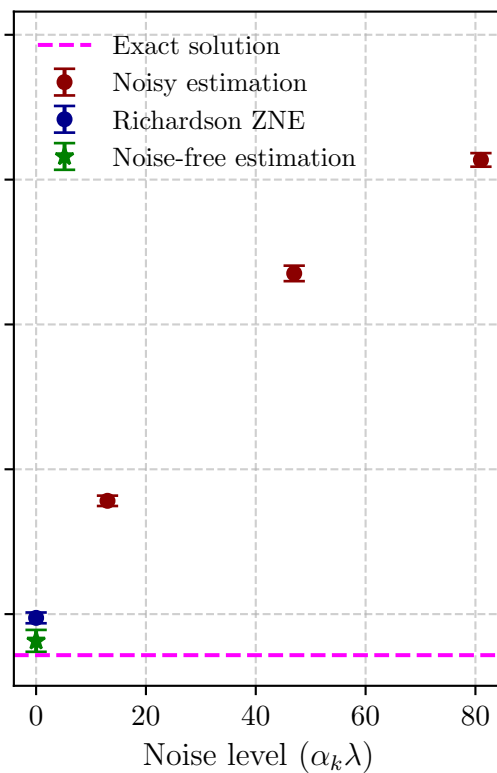


| Description | Simulation            | Mean $\pm$ Std         |
|-------------|-----------------------|------------------------|
| xy-ric2     | vqe                   | -6.43550 $\pm$ 0.07145 |
| xy-ric2     | redundant (noise=13)  | -6.43550 $\pm$ 0.07145 |
| xy-ric2     | redundant (noise=47)  | -3.29514 $\pm$ 0.10699 |
| xy-ric2     | zne                   | -7.63623 $\pm$ 0.07321 |
| xy-ric2     | noiseoff              | -8.36934 $\pm$ 0.15125 |
|             |                       |                        |
| xy-ric3     | vqe                   | -6.43550 $\pm$ 0.07145 |
| xy-ric3     | redundant (noise=13)  | -6.43550 $\pm$ 0.07145 |
| xy-ric3     | redundant (noise=47)  | -3.29514 $\pm$ 0.10699 |
| xy-ric3     | redundant (noise=81)  | -1.72787 $\pm$ 0.09369 |
| xy-ric3     | zne                   | -8.05195 $\pm$ 0.07442 |
| xy-ric3     | noiseoff              | -8.36934 $\pm$ 0.15125 |
|             |                       |                        |
| xy-ric4     | vqe                   | -6.43550 $\pm$ 0.07145 |
| xy-ric4     | redundant (noise=13)  | -6.43550 $\pm$ 0.07145 |
| xy-ric4     | redundant (noise=47)  | -3.29514 $\pm$ 0.10699 |
| xy-ric4     | redundant (noise=81)  | -1.72787 $\pm$ 0.09369 |
| xy-ric4     | redundant (noise=115) | -0.92017 $\pm$ 0.07099 |
| xy-ric4     | zne                   | -8.22268 $\pm$ 0.07642 |
| xy-ric4     | noiseoff              | -8.36934 $\pm$ 0.15125 |
|             |                       |                        |
| xy-ric5     | vqe                   | -6.43550 $\pm$ 0.07145 |
| xy-ric5     | redundant (noise=13)  | -6.43550 $\pm$ 0.07145 |
| xy-ric5     | redundant (noise=47)  | -3.29514 $\pm$ 0.10699 |
| xy-ric5     | redundant (noise=81)  | -1.72787 $\pm$ 0.09369 |
| xy-ric5     | redundant (noise=115) | -0.92017 $\pm$ 0.07099 |
| xy-ric5     | redundant (noise=149) | -0.49237 $\pm$ 0.05256 |
| xy-ric5     | zne                   | -8.29966 $\pm$ 0.07818 |
| xy-ric5     | noiseoff              | -8.36934 $\pm$ 0.15125 |
|             |                       |                        |
| xy-ric6     | vqe                   | -6.43550 $\pm$ 0.07145 |
| xy-ric6     | redundant (noise=13)  | -6.43550 $\pm$ 0.07145 |
| xy-ric6     | redundant (noise=47)  | -3.29514 $\pm$ 0.10699 |
| xy-ric6     | redundant (noise=81)  | -1.72787 $\pm$ 0.09369 |
| xy-ric6     | redundant (noise=115) | -0.92017 $\pm$ 0.07099 |
| xy-ric6     | redundant (noise=149) | -0.49237 $\pm$ 0.05256 |
| xy-ric6     | redundant (noise=183) | -0.26091 $\pm$ 0.03978 |
| xy-ric6     | zne                   | -8.33662 $\pm$ 0.07942 |
| xy-ric6     | noiseoff              | -8.36934 $\pm$ 0.15125 |
|             |                       |                        |
| xy-ric7     | vqe                   | -6.43550 $\pm$ 0.07145 |
| xy-ric7     | redundant (noise=13)  | -6.43550 $\pm$ 0.07145 |
| xy-ric7     | redundant (noise=47)  | -3.29514 $\pm$ 0.10699 |
| xy-ric7     | redundant (noise=81)  | -1.72787 $\pm$ 0.09369 |
| xy-ric7     | redundant (noise=115) | -0.92017 $\pm$ 0.07099 |
| xy-ric7     | redundant (noise=149) | -0.49237 $\pm$ 0.05256 |
| xy-ric7     | redundant (noise=183) | -0.26091 $\pm$ 0.03978 |
| xy-ric7     | redundant (noise=217) | -0.13396 $\pm$ 0.03105 |
| xy-ric7     | zne                   | -8.35520 $\pm$ 0.08022 |
| xy-ric7     | noiseoff              | -8.36934 $\pm$ 0.15125 |
|             |                       |                        |

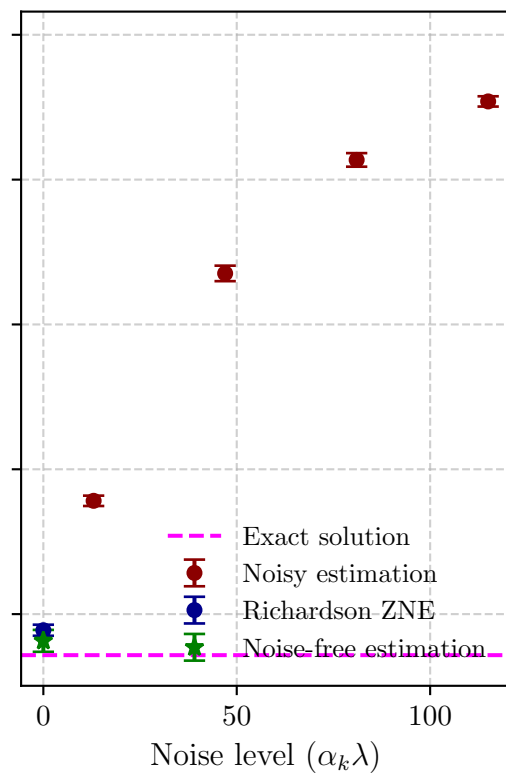
XY ansatz ric-2



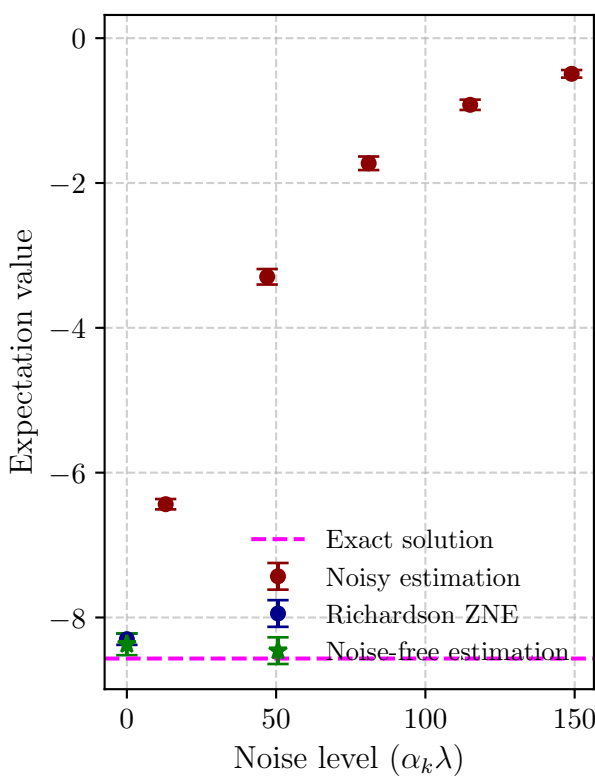
XY ansatz ric-3



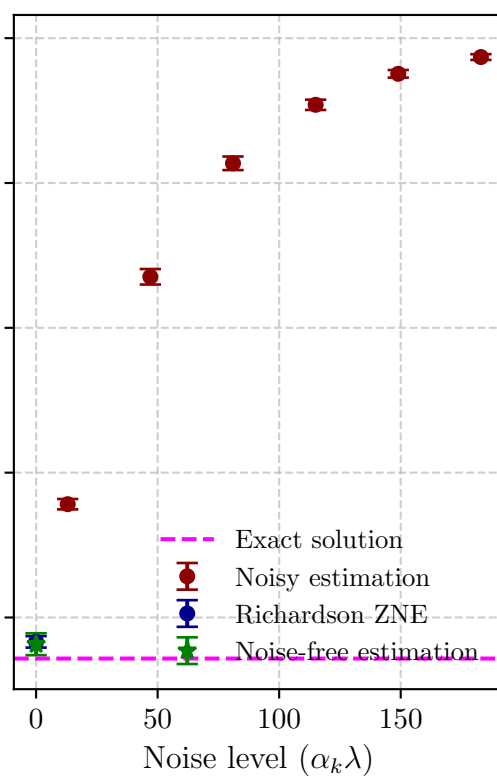
XY ansatz ric-4



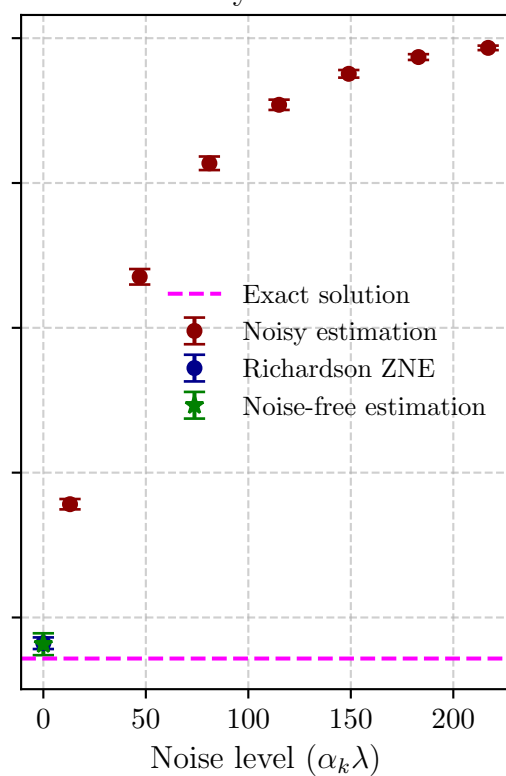
XY ansatz ric-5



XY ansatz ric-6



xy-ric7



# XY VQE CONFIGURATION

| Parameter                                | Value                               |
|------------------------------------------|-------------------------------------|
| run                                      | vqe                                 |
| nqubits                                  | 7                                   |
| state                                    | dmatrix                             |
| output.file_name_prefix                  | xy_noisey_time_evo                  |
| output.draw.status                       | True                                |
| output.draw.fig_dpi                      | 100                                 |
| output.draw.type                         | png                                 |
| observable.def                           | ising                               |
| observable.coefficients.cn               | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| observable.coefficients.bn               | [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0] |
| observable.coefficients.r                | 1                                   |
| ansatz.layer                             | 30                                  |
| ansatz.gateset                           | 1                                   |
| ansatz.ugate.type                        | xy-iss                              |
| ansatz.ugate.coefficients.cn             | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| ansatz.ugate.coefficients.bn             | [0, 0, 0, 0, 0, 0, 0]               |
| ansatz.ugate.coefficients.r              | 0                                   |
| ansatz.ugate.time.min                    | 0.0                                 |
| ansatz.ugate.time.max                    | 10.0                                |
| vqe.iteration                            | 10                                  |
| vqe.optimization.status                  | True                                |
| vqe.optimization.algorithm               | SLSQP                               |
| vqe.optimization.constraint              | True                                |
| init_param.value                         | random                              |
| noise_profile.status                     | True                                |
| noise_profile.type                       | depolarizing                        |
| noise_profile.noise_prob                 | [0.001, 0.001, 0.001, 0.001]        |
| noise_profile.noise_on_init_param.status | False                               |
| noise_profile.noise_on_init_param.value  | 0                                   |
| redundant.identity_factors               | [[0, 0, 0, 0], [1, 1, 0, 0]]        |
| zne.method                               | richardson                          |
| zne.degree                               | 1                                   |
| zne.sampling                             | default                             |
| zne.data_points                          |                                     |